

BUSINESS ANALYTICS PROJECT REPORT

Starbucks Beverage Analytics Dashboard

An interactive Power BI executive dashboard for beverage nutrition and global store presence

Industry	Domain	Platform
Food & Beverage	Business Intelligence	Microsoft Power BI

Prepared for GitHub portfolio and recruiter review

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This PDF uses a fixed page layout for a concise portfolio case study. The Markdown companion provides GitHub navigation links.

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Scope note. *The supplied project brief identifies three source files and a dashboard image, but those attachments were unavailable in the handoff workspace. The report preserves the declared project scope and intentionally does not state unverified counts, values, or visual observations.*

1. Executive Summary

The Starbucks Beverage Analytics Dashboard is an interactive Power BI solution for exploring beverage nutrition and global store presence. It brings together beverage-level attributes - calories, sugar, caffeine, protein, and category - with geographic footprint information so stakeholders can move from static product tables to a focused executive view.

The dashboard was created to reduce the effort needed to compare categories, surface high-caffeine items, and frame product discussion alongside location coverage. KPI cards, category comparisons, rankings, a geographic view, and slicers make the report useful to business managers, marketing teams, product managers, nutrition analysts, and BI teams.

Its value is decision support: standardised definitions and filter-aware calculations make recurring questions quicker to answer, while the model retains a traceable path back to the cleaned source. The result is a portfolio-quality example of source preparation, data modeling, DAX, dashboard design, and business storytelling.

Business value

- Creates one analytical experience for product and footprint questions.
- Makes category variation visible without losing beverage-level detail.
- Supports an auditable path from KPI to underlying records.
- Provides a maintainable foundation for governed refresh and richer commercial data.

Outcome: *The project demonstrates a complete BI workflow - Power Query preparation, relational modeling, DAX measures, interactive design, and business documentation.*

2. Business Problem and Objectives

Starbucks operates across a broad beverage portfolio and a global store network. Product information is commonly stored in static tables while location data is reviewed separately; that fragmentation slows cross-category comparisons and makes it harder to tell a clear business story.

Nutrition analysis matters because calories, sugar, caffeine, and protein are material product attributes for menu planning, customer communication, and targeted marketing. Store presence is a footprint signal that can support market-coverage conversations. Power BI improves decision-making by applying consistent measures and maintaining filter context in a single navigable interface.

Project objectives

- Create one interactive view of beverage nutrition and store presence.
- Standardise category, numeric, and geography fields before analysis.
- Publish consistent KPIs for beverage count and average nutrition measures.
- Enable category comparison and filter-aware caffeine ranking.
- Support country or regional exploration at the source grain.
- Design a clear executive hierarchy with accessible interactions.
- Document measures and model rules for maintainability.

3. Dataset Overview and Preparation

The project brief identifies three CSV/Excel datasets plus a dashboard image. The intended scope combines beverage nutrition records with Starbucks location/store information and a supporting reference or mapping source. Exact rows, fields, time period, and source grain must be confirmed from the original uploads before publication.

Domain	Expected grain	Illustrative attributes	Use
Beverage nutrition	One row per drink/menu item	Name, category, calories, sugar, caffeine, protein	Comparisons and ranking
Store presence	Country/market or store-level row	Country/region, store count or store ID	Footprint view
Reference	One mapped category/location	Standard labels, sort order, mapping	Consistent reporting

Power Query preparation

- Profile headers, duplicates, nulls, unexpected categories, and record counts.
- Promote headers; trim whitespace; standardise text casing and location labels.
- Assign explicit numeric, text, and geography-compatible data types.
- Treat blanks and invalid nutrition values as exceptions, not automatic zeroes.
- Retain an audit step showing rows before and after transformations.

Validation required: Confirm serving basis and units, source date, coverage, and whether location data is pre-aggregated. Those four facts govern interpretation of averages and totals.

4. Data Model and Dashboard Design

A small star schema is the preferred model. Beverage Nutrition remains at beverage grain and Store Presence remains at its native geographic grain. Shared dimensions should be introduced only where a stable key exists; independent subject areas do not need an artificial relationship to coexist on an executive report page.

Principle	Implementation rationale
Explicit grain	Prevents duplicated beverage averages or incorrect sums of pre-aggregated store totals.
One-to-many relationships	Dimensions filter facts without ambiguous bidirectional paths.
Stable categories	A category dimension can control display labels and sort order.
Separate subject areas	Protects nutrition calculations from accidental geographic duplication.

Dashboard sections

Section	Business value
KPI cards	Quick baseline for count and average calories, sugar, and caffeine.
Global presence	Shows recorded footprint concentration and coverage.
Beverage distribution	Frames assortment composition before interpreting averages.
Nutrition comparisons	Identifies categories warranting deeper review.
Top caffeine drinks	Turns category patterns into beverage-level detail.
Interactive filters	Lets users answer focused questions without rebuilding analysis.

Interaction design

Category selections should update applicable nutrition cards and rankings. Geography controls should affect store visuals only unless a validated common dimension supports a wider interaction. Visible selection states, reset controls, concise tooltips, and consistent units improve executive usability.

5. DAX Measures

DAX ensures that cards, charts, tooltips, and filters use the same calculation logic. The following examples use placeholder table and column names and should be matched to the final model.

```
Total Beverage Count = DISTINCTCOUNT ( 'Beverage Nutrition'[Beverage Name] )
Average Calories = AVERAGE ( 'Beverage Nutrition'[Calories] )
Average Sugar (g) = AVERAGE ( 'Beverage Nutrition'[Sugar (g)] )
Average Caffeine (mg) = AVERAGE ( 'Beverage Nutrition'[Caffeine (mg)] )
Caffeine Rank = RANKX ( ALLSELECTED ( 'Beverage Nutrition'[Beverage Name] ), CALCULATE ( MAX (
'Beverage Nutrition'[Caffeine (mg)] ) ), , DESC, DENSE )
Top 5 Caffeine Flag = IF ( [Caffeine Rank] <= 5, 1, 0 )
```

Measure	Business purpose	Design consideration
Total Beverage Count	Visible assortment size	Use a stable key when beverage names repeat.
Average Calories	Comparable energy indicator	Verify equivalent serving basis.
Average Sugar (g)	Category sugar pattern	Display units; exclude invalid values.
Average Caffeine (mg)	Stimulant-content comparison	Document blank/decaf handling.
Rank / Top 5 Flag	Context-aware top items	ALLSELECTED preserves user intent.

Testing rule: Validate each measure under slicers, cross-highlights, and no-selection states. A correct total at report level can still be misleading in a visual if its filter behavior is not understood.

6. Key Insights

The report is designed to surface the following questions and directional insights. They are not expressed as numerical findings because the original sources and dashboard image were unavailable for verification.

- Category averages should be read with beverage counts and distribution, not isolated extremes.
- High-calorie and high-sugar patterns may diverge; the dashboard makes the trade-off inspectable.
- Caffeine leadership is most useful after category filtering, rather than from a portfolio average alone.
- Small categories can have volatile averages, so count context is essential.
- Serving-size consistency is a prerequisite for defensible nutrition comparisons.
- Top-caffeine rankings become more actionable when they retain the selected category context.
- Filterable visuals enable focused comparisons instead of static, one-size-fits-all benchmarks.
- Store presence is a footprint signal - not a proxy for revenue, demand, or profitability.
- Country totals need source-grain validation to avoid duplicate aggregation.
- Geographic concentration can prompt questions about market maturity and reporting needs.
- The dashboard connects product transparency with commercial context without implying causality.
- Clear units, definitions, and tooltips are core to executive trust.

7. Business Recommendations

- Establish a standard serving-size basis or label serving differences on every nutrition visual.
- Create a lower-calorie / lower-sugar exploration path using thresholds approved by nutrition stakeholders.
- Use caffeine ranking in product information workflows with suitable health disclaimers.
- Add a calories-versus-sugar scatter plot to reveal trade-offs hidden by averages.
- Track the share of beverages meeting formally agreed product-nutrition criteria.
- Display source version and refresh date for nutrition and store information.
- Segment store views by region or market maturity only when governed dimensions are available.
- Publish a data dictionary and metric definitions beside the report.
- Use stakeholder feedback and usage telemetry to refine the visual set.
- Pair store footprint analysis with sales, customer, and profitability data before market decisions.

Recommended next release: Add a governed refresh process and a one-page data dictionary. These changes improve trust and handover value more than adding additional charts.

Decision framing

The dashboard supports exploratory questions and should guide, not replace, product, health, or market decisions. Any action should be tested against source recency, business constraints, and relevant stakeholder review.

8. Challenges and Future Enhancements

Challenges addressed

Challenge	Practical response
Inconsistent labels and types	Standardise in Power Query and retain an exceptions review.
Missing or ambiguous values	Separate blanks from genuine zeroes; document exclusion rules.
Different source grains	Define grain before creating relationships or aggregations.
Visual overload	Use hierarchy: KPIs, comparisons, then ranked detail.
Context-sensitive DAX	Test slicers, cross-highlights, and no-selection states.
Performance and usability	Limit unused columns and test the published report.

Future enhancements

- Move to SQL Server or Azure SQL for governed storage and incremental refresh.
- Connect an approved API or enterprise product-information feed where licensing permits.
- Publish to Power BI Service with scheduled refresh and semantic-model governance.
- Add mobile-first layouts and an executive summary page.
- Introduce historical snapshots to analyse assortment and footprint change.
- Combine nutrition with sales, transactions, or customers for performance analysis.
- Explore forecasting and segmentation when time-series or customer data exists.
- Apply row-level security when roles and geographic ownership are defined.

9. Skills, Tools, and Repository Structure

Skills demonstrated

Business intelligence; Power BI; Power Query; DAX; data modeling; data cleaning; KPI design; dashboard design; data visualization; business analysis; analytical thinking; stakeholder storytelling; problem solving; and technical documentation.

Tools used

Tool	Purpose
Microsoft Power BI	Semantic model, interactive report, and publishing workflow.
Power Query	Source ingestion, profiling, cleaning, typing, and transformation.
DAX	Filter-aware KPIs, averages, rankings, and visual logic.
Excel / CSV	Source-data exchange and exploratory validation.
GitHub	Versioned portfolio documentation, assets, and reproducibility.

10. Conclusion and Appendix

This project turns two related but distinct lenses - beverage nutrition and store footprint - into a usable executive analytics experience. It demonstrates how disciplined preparation, explicit model grain, filter-aware DAX, and purposeful visual design make product information easier to explore and communicate. With validated sources, documented refresh rules, and the recommended enhancements, the dashboard can mature from a portfolio case study into a governed decision-support asset.

Appendix A - Data dictionary template

Field	Definition	Validation rule
Beverage Name	Published beverage/menu label	Non-blank; duplicates reviewed.
Beverage Category	Standardised product group	Mapped to approved category list.
Calories / Sugar / Caffeine / Protein	Nutrition values per documented serving	Non-negative; units and serving basis shown.
Country / Region	Geographic reporting label	Standardised against mapping table.
Store Count / Store ID	Presence measure at declared grain	Aggregation matches source grain.

Appendix B - Assumptions and limitations

- Nutrition values are interpreted only at the source-supplied serving basis.
- Store-data timing, coverage, and grain must be confirmed before market claims.
- The dashboard does not measure sales, customer preference, margin, demand, or health outcomes.
- No numerical insight here is source-verified until the original uploads are restored and reconciled.

Appendix C - Quality assurance checklist

- Reconcile raw-to-curated row counts and validate units.
- Document null, zero, duplicate, and outlier rules.
- Test relationships, filter direction, DAX measures, titles, tooltips, and accessibility.
- Show source/version date in the published report.